WORKED SOLUTION

STEP II 1988, Question 12

Mechanics • Falling Chains

STEP II 1988, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

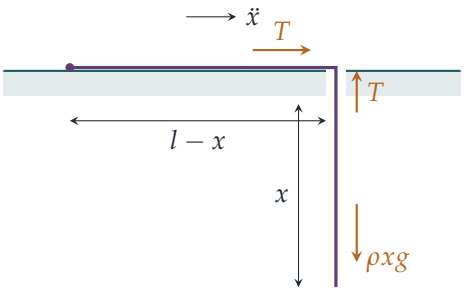
step.maths.org/questions/88-s2-q12

The key idea

Because the string is inextensible, every bit of it moves with the same speed, so the whole string behaves like one body whose driving force is the weight of the part that hangs below the table. That weight grows as more string goes through the hole, which gives a simple linear differential equation in the hanging length. Its solution is a hyperbolic cosine, and the time we want comes from inverting it.

Solution

There is no friction anywhere: the table is smooth, the hole is small and smooth (so it changes the string's direction without exerting any force along it), and the string has mass ρ per unit length. Let x be the length hanging below the table at time t, so that the length lying on the table is $l - x$. Initially $x = l - d$ and $\dot{x} = 0$, and the end reaches the top of the hole when $x = l$. Since the table is higher than l, the hanging part never touches the floor before then.



Let T be the tension in the string at the hole. As the hole is smooth, the tension has the same value T just above and just below it.

The part on the table. It has mass $\rho(l - x)$ and moves horizontally towards the hole with acceleration $\ddot{x}$ (inextensibility: it moves as fast as the hanging part). The only

horizontal force on it is the tension T, since the table is smooth. So

$$\rho(l-x)\,\ddot{x} = T.$$

The hanging part. It has mass ρx and accelerates downwards at $\ddot{x}$, under its weight $\rho x g$ downwards and the tension T upwards:

$$\rho x\,\ddot{x} = \rho x g - T.$$

Adding the two equations eliminates T:

$$\rho l\,\ddot{x} = \rho g x, \qquad \text{that is} \qquad \ddot{x} = n^2 x, \quad n = \sqrt{\frac{g}{l}}.$$

Ask yourself The amount of string on each side of the hole keeps changing. Is it really all right to use $F = ma$ for each piece?

Yes. Each equation is applied at one instant to the particular piece of string that is on that side of the hole at that instant; the string moves as a whole with a single speed $\dot{x}$, so the rate of change of momentum of the whole string is $\rho l\ddot{x}$, and the only external force with a component along the direction of motion is the weight $\rho x g$ of the hanging part. (The reaction of the table balances the weight of the horizontal part, and the hole only turns the string.) So the combined equation $\rho l\ddot{x} = \rho g x$ is just Newton's second law for the whole string. An energy check below confirms it.

The general solution of $\ddot{x} = n^2 x$ is $x = A\cosh nt + B\sinh nt$. From $\dot{x} = 0$ at $t = 0$ we get $B = 0$, and from $x = l - d$ at $t = 0$ we get $A = l - d$:

$$x = (l-d)\cosh nt.$$

The end reaches the top of the hole when $x = l$, so at the time t_1 with

$$\cosh nt_1 = \frac{l}{l-d}.$$

Since $l/(l-d) > 1$ this has exactly one positive solution, and using $\operatorname{arcosh} u = \ln(u + \sqrt{u^2-1})$,

$$\sqrt{u^2-1} = \frac{\sqrt{l^2-(l-d)^2}}{l-d} = \frac{\sqrt{2ld-d^2}}{l-d},$$

so the time taken is

$$\boxed{t_1 = \sqrt{\frac{l}{g}}\,\operatorname{arcosh}\!\left(\frac{l}{l-d}\right) = \sqrt{\frac{l}{g}}\,\ln\left(\frac{l+\sqrt{2ld-d^2}}{l-d}\right).}$$

Checks

- **Energy.** Take the table top as the zero of potential energy. The hanging part has its centre of mass at depth $x/2$, so its potential energy is $-\frac{1}{2}\rho g x^2$. Conservation of

energy from rest gives $\frac{1}{2}\rho l\dot{x}^2 = \frac{1}{2}\rho g\big(x^2 - (l-d)^2\big)$, and differentiating with respect to t gives $l\ddot{x} = gx$ again.

- **Small d.** If d is tiny, almost all the string hangs and it should fall with acceleration close to g, taking time about $\sqrt{2d/g}$. Indeed, for small d, $\cosh nt_1 = 1 + \frac{d}{l} + \cdots$ and $\cosh nt_1 \approx 1 + \frac{1}{2}n^2t_1^2$ give $t_1^2 \approx 2d/(n^2l) = 2d/g$.
- **$d \to l$.** If almost nothing hangs, the string barely starts, and $t_1 \to \infty$, as the logarithm shows.
- **Numerically.** With $l = 2\,\mathrm{m}$, $d = 1.5\,\mathrm{m}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$ the formula gives $t_1 \approx 0.932\,\mathrm{s}$, which matches a step-by-step numerical integration of the motion.

Common slips

- Writing $\rho x\ddot{x} = \rho xg$ for the hanging part alone, forgetting that it has to drag the string on the table (equivalently, using the wrong mass: the mass being accelerated is the whole string, ρl).
- Trying a solution of the form $x = A\cos nt$: the equation is $\ddot{x} = +n^2x$, so the motion grows exponentially rather than oscillating.
- Setting $x = d$ instead of $x = l$ for the finishing condition: the end reaches the hole when the whole length l is hanging.

Beyond STEP

This is a variable-mass-distribution problem: the mass is fixed, but the part feeling gravity changes with position, leading to a linear equation whose solutions are exponentials. If the string fell off a frictionless table edge with a pile of string being picked up from rest, momentum would be lost in inelastic "jerks" and energy would not be conserved; telling those cases apart is a classic theme in undergraduate mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a careful treatment of systems with moving mass and of chains and ropes.

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